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Intuitive-K-prototypes: A mixed data clustering algorithm with intuitionistic distribution centroid

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ABSTRACT

Data sets are usually mixed with numerical and categorical attributes in the real world. Data mining of mixed data makes a lot of sense. This paper proposes an Intuitive-K-prototypes clustering algorithm with improved prototype representation and attribute weights. The proposed algorithm defines intuitionistic distribution centroid for categorical attributes. In our approach, a heuristic search for initial prototypes is performed. Then, we combine the mean of numerical attributes and intuitionistic distribution centroid to represent the cluster prototype. In addition, intra-cluster complexity and inter-cluster similarity are used to adjust attribute weights, with higher priority given to those with lower complexity and similarity. The membership and non-membership distance are calculated using the intuitionistic distribution centroid. These distances are then combined parametrically to obtain the composite distance. The algorithm is judged for its clustering effectiveness on the real UCI data set, and the results show that the proposed algorithm outperforms the traditional clustering algorithm in most cases.

1. Introduction

Clustering is an unsupervised machine learning technique for grouping unlabeled data into clusters containing data points that are "similar" to each other but "different" from data points in other clusters [1,2]. This method helps us discover structures and patterns in the data and is often used in many fields, such as data analysis, market segmentation, social network analysis, and image segmentation. Many data sets will be mixed with numerical and categorical attributes in the real world. Still, many clustering algorithms can only deal with data containing numerical(FC-K-means [3]) or categorical(DP-k-modes [4]) features [5,6].

Numerical attributes are generally real-valued, with typical distance metrics such as Manhattan distance, Euclidean distance, Chebyshev distance, etc. On the other hand, categorical attributes have different values that represent different categories and distinguish objects from other properties. Therefore, it is not recommended to compute the distance between different values of categorical attributes directly. Thus, proposing a precise distance measure for categorical attributes is crucial.

Some works have been published on mixed data clustering. Huang [7] proposed the K-prototypes algorithm for mixed data by combining the K-means and K-modes algorithms. However, the K-prototypes clustering algorithm must be revised to effectively explain categorical attribute values distribution.

So, the distribution of categorical attribute values in the cluster was considered in [8,9]. Ji and Bai [10] proposed the IK-prototypes algorithm using the distribution centroid to represent categorical attributes of cluster centers, similar to [8,9]. Sangam and Om [11] proposed the EBK-prototypes algorithm, which considers the distribution of categorical attribute values within and between clusters to derive the distance. Li et al. [12] proposed the MCFCIW algorithm by filtering the distribution centroid.

There are also other clustering methods in recent years, as shown below. Ahmad and Khan [13] proposed the InitKmix algorithm, which separately clusters each feature and merges for the final cluster to create the initial partitions of other algorithms. Li et al. [14] expanded the space structure representation scheme for categorical data to mixed data and proposed the SBM and Ap-SBM frameworks for clustering. Rezaei and Daneshpour [15] proposed the SFK-prototypes algorithm, which defines the concept of similar attributes. This method considers the two criteria of distance and the number of similar attributes when clustering.

Then, a new mixed data clustering algorithm is proposed in this paper. Its main advantages are listed below.

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1. We propose a reasonable search method for initial prototypes based on the approximate farthest distance of the data. Compared with most algorithms that randomly select initial prototypes, this method can provide initial prototypes with a more uniform distribution for clustering and, in most cases, obtain a stable clustering effect.
2. We apply the intuitionistic fuzzy set model to represent the distribution of categorical attribute values and propose the concept of the intuitionistic distribution centroid, which can more accurately represent the distribution of attribute values.

bước này sẽ chọn centroids ban đầu cho từng cụm, thay vì random như ở K-prototypes

3. In the attribute weights iteration, we construct attribute weights by considering the complexity of intra-cluster attributes and the similarity of inter-cluster attributes. Different from [12], we propose some new complexity and similarity calculation methods that perform well in experiments.

Based on the above strategies, we propose the Intuitive-K-prototypes algorithm. This algorithm aims to record the distribution of categorical attribute values more accurately and quantify attribute weights during iteration. It should provide reliable clustering with higher performance and lower complexity.

The rest of this paper is organized as follows: In the next section, we outline the mixed data clustering-related work about three essential aspects. Section 3 introduces related concepts and the initial prototype selection method. Then, in Section 4, we describe the Intuitive-K-prototypes algorithm. The experimental results and analysis are given in Section 5. Finally, we summarize the content of this paper and outline future research directions.

2. Related work

The three essential aspects of the mixed data clustering algorithm are selecting initial prototypes, representing the center, and updating the weight strategy. Some related work is introduced below.

Selecting the appropriate initial prototypes is crucial in achieving a successful clustering outcome. There are some ways to choose initial centers. In [16], the algorithm requires determining the candidate medoids subsets and calculating the distance matrix, then using both to incrementally increase the number of clusters and new medoids from 2 to K. In [17], the dispersion degree is defined based on entropy. The largest and smallest dispersion objects are selected as the initial cluster centers. In [18,19], it is considered that the perfect prototype has a high density and a considerable distance from other data objects with higher density. The time complexity of all the above methods is $O\left(n^2\right)$. In the next section, we provide a prototype selection algorithm with a time complexity of $O(n)$.

For cluster centers of mixed data, the most important thing is the representation of categorical attributes, generally described by medoids, modes, and distribution centroids. In [20,21], the center is represented by the medoid of the cluster that has the smallest sum of distances to each object. However, finding the medoid is time-consuming. In [7], the prototype of mixed attribute clustering is proposed as a combination of the mean of the numerical attribute in the cluster and the mode of the categorical attribute. As for the representation of the categorical attribute part, much information in the cluster will be lost. In [12], the distribution centroid is representing categorical attributes. The distribution centroid is more accurate than that of medoids and prototypes. Its construction depends on the information in the cluster, but does not pay attention to the distribution of attribute values outside the cluster. It has no way of distinguishing between two different attribute values if they have the same proportion in one cluster and different proportion in the other. This paper overcomes the shortcoming.

For strategies for adjusting weights for mixed data clustering, K-prototypes [7] use a single weighting factor to balance numerical and categorical attributes. It does not clearly state the importance of each attribute. For the weight update in [22], the attribute weights consider the distance of each object from the cluster center. The updated weights in [12] consider the information between cluster centers, which is more convenient to calculate. In [23], entropy and Jensen–Shannon divergence concepts are used to exploit the inter-attribute information of categorical–categorical and categorical–numerical attributes, respectively. Next, we will explain the idea of this paper.

3. Initial prototypes selection

Our discussions and analysis are based on the following: The mixed data set $X = \left\{x_1, x_2, \ldots, x_n\right\}$ with m attributes $\left\{A_1, A_2, \ldots, A_m\right\}$ is described as follows. The domain of each attribute $A_j\,(1 \le j \le m)$ is defined as $Dom\left(A_j\right)$. The m attributes are divided into m_n numerical attributes $\left\{A^n_{1_n}, A^n_{2_n}, \ldots, A^n_{m_n}\right\}$ and m_c categorical attributes $\left\{A^c_{1_c}, A^c_{2_c}, \ldots, A^c_{m_c}\right\}$. Objects in a mixed data set can be represented as vectors $x_i = \left(x_{i1}, x_{i2}, \ldots, x_{im}\right)\ \left(x_{ij} \in Dom\left(A_j\right)\right)$.

3.1. Initial prototypes selection method

Most algorithms randomize the choice of initial prototypes, which results in a random effect. The prototypes should align with the data division to ensure the effectiveness of algorithm. Prototypes selected by constructing local densities in [24] are more representative, but the method is highly complex and does not involve updating prototypes. In this paper, the prototypes will fit the clustering better in the iterative update, so we just need to get the uniformly dispersed initial prototypes.

The farthest distance $d_{\max}$ in the data set X is one of the data distribution characteristics. However, the time complexity of the computation is $O\left(n^2\right)$. We reduce the time complexity by applying the approximation d_{approx} of $d_{\max}$.

In the process of selecting prototypes, the specific form of the distance is:

$$d\left(x_i, x_l\right) = \sum_{j_1=1_n}^{m_n} \left|x_{ij_1} - x_{lj_1}\right| + \sum_{j_2=1_c}^{m_c} \delta\left(x_{ij_2}, x_{lj_2}\right) \tag{1}$$

where, $\delta\left(x_{ij_2}, x_{lj_2}\right) = \begin{cases} 1 & x_{ij_2} \ne x_{lj_2} \\ 0 & x_{ij_2} = x_{lj_2} \end{cases}$.

Suppose that we want to divide the reasonably distributed data set X into k clusters; we consider that the distances between the centers of k clusters are greater than or equal to $d_\lambda = \frac{(\lambda d_{approx})}{k}\ (0 < \lambda \le 1)$. With this idea, we can obtain initial prototypes set with uniform distribution. The process is as follows:

Step 1: Randomly select an object x^0 in the data set X.

Step 2: Find the x^1 with the largest distance from x^0 in X.

Step 3: Find the object x^2 with the largest distance from x^1 in X, we get the approximately maximum distance $d_{approx} = d\left(x^1, x^2\right)$.

Step 4: Find objects in X whose distances to elements in the prototype set Q are all greater than d_λ until $|Q| = k$.

Algorithm 1 Initial prototype selection

Input: mixed data sets $X = \left\{x_1, x_2, ..., x_n\right\}$, number of clusters k, distance parameter λ.
Output: initial set of prototypes $Q = \left\{q^1, q^2, ..., q^k\right\}$.
Initial: prototypes set $Q = \emptyset$, Choose a point x^0 at random in X.

```
for i = 1 to n do
    calculate the distance between x_i and x^0 by Eq. (1).
    select the object x^1 with the largest distance.
end for
Q = Q ∪ {x^1}
for i = 1 to n do
    calculate the distance between x_i and x^1 by Eq. (1).
    select the object x^2 with the largest distance.
```

10: **end for**
12: $Q = Q \cup \{x^2\}$
13: **if** $|Q| < k$ **then**
14: **for** $i = 1$ to n **do**
15: **for** $j = 1$ to $|Q|$ **do**
16: calculate the distance between x_i and $x^j \in Q$ by Eq. (1).
17: **end for**
18: select x^i whose distances satisfy $d\left(x^i, x^j\right) \geq d_\lambda (x^j \in Q)$.
19: **end for**
20: $Q = Q \cup \{x^i\}$
21: **end if**
22: **if** $|Q| < k$ **then**
23: **repeat** step.14 – step.19 until $|Q| = k$.
24: **end if**
25: **return** initial set of prototypes $Q = \{q^1, q^2, ..., q^k\}$.

3.2. Algorithm time complexity analysis

We can see that the time complexity of the Algorithm 1 is mainly in a couple of loops; the time complexity of this algorithm is $O\left(2n + \frac{(k-2)(k+1)}{2}n\right)$ of which $k \ll n$. We can consider the time complexity of this algorithm as linear $O(n)$, and it has ideal computing efficiency.

To demonstrate the feasibility of the method, we selected two attributes of the Iris data set to apply the above process and obtained three reasonable initial prototypes in Figs. 1–3.

4. The intuitive-K-protoypes algorithm with intuitionistic distribution centroid

In this section, we aim to measure the similarity between categorical attributes more accurately and detailedly. The cluster prototypes Q_l consist of numerical parts Q_{l_n} and categorical parts Q_{l_c}. Numerical parts Q_{l_n} use the mean value of attributes in the cluster. Categorical parts Q_{l_c} are discussed in this section.

4.1. Intuitionistic distribution centroid

Suppose that k clusters $\{C_1, C_2, \ldots, C_k\}$ are obtained after applying Eq. (1) for assigning objects to the nearest prototype. After obtaining k clusters, the intuitionistic distribution centroid is updated as follows.

Definition 1 (*Intuitionistic Distribution Centroid*)**.** Assume the intuitionistic distribution centroid of cluster C_l is $Q_{l_c} = \left\{q_{l1_c}, q_{l2_c}, \ldots, q_{lm_c}\right\}$, then the categorical attributes for q_{lj_c} can be defined with formulas (2)–(4):

$$q_{lj_c} = \left\{\left\langle a^1_{j_c}, \mu^1_{lj_c}, v^1_{lj_c}\right\rangle, \left\langle a^2_{j_c}, \mu^2_{lj_c}, v^2_{lj_c}\right\rangle, \ldots, \left\langle a^t_{j_c}, \mu^t_{lj_c}, v^t_{lj_c}\right\rangle\right\} (1_c \leq j_c \leq m_c) \quad (2)$$

$$\mu^k_{lj_c} = \left|\left[a^k_{j_c}\right] \cap C_l\right|^2 / \left|\left[a^k_{j_c}\right]\right| |C_l| \, (1 \leq k \leq t) \quad (3)$$

$$v^k_{lj_c} = \sum_{s \neq l} \left(\left|\left[a^k_{j_c}\right] \cap C_s\right| / \left|\left[a^k_{j_c}\right]\right|\right)^2 (1 \leq k \leq t) \quad (4)$$

where, t is the number of attribute values of the j_c categorical attribute, C_l is the set of objects in cluster l, $\left[a^k_{j_c}\right] = \left\{x_i : x_{ij} = a^k_{j_c}\right\}$ is consisting of objects which j_c categorical attribute is $a^k_{j_c}$, $|C_l|$ and $\left|\left[a^k_{j^c}\right]\right|$ are the number of elements in the corresponding set.

$\mu^k_{lj_c}$ and $v^k_{lj_c}$ represent the membership and non-membership of the attribute value $a^k_{j_c}$ of the j_c categorical attribute in cluster C_l.

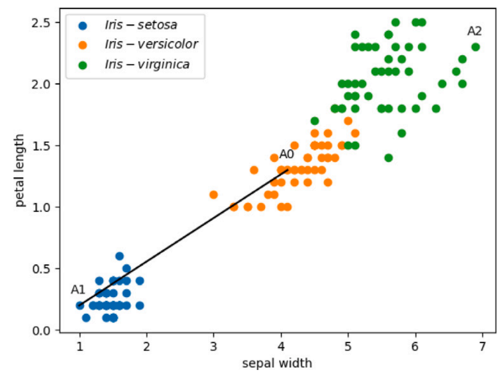

Fig. 1. Randomly select an object $A0$ and Find the $A1$ with the largest distance from $A0$.

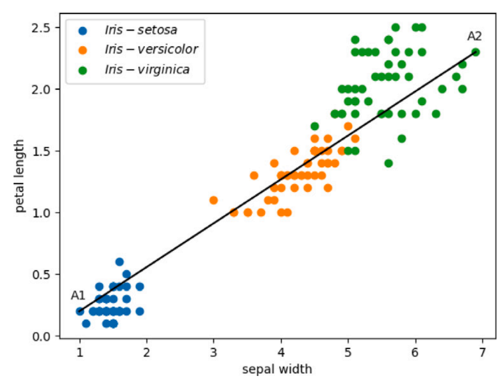

Fig. 2. Find the object $A2$ with the largest distance from $A1$ and get the approximately maximum distance $d_{approx} = d(A1, A2)$.

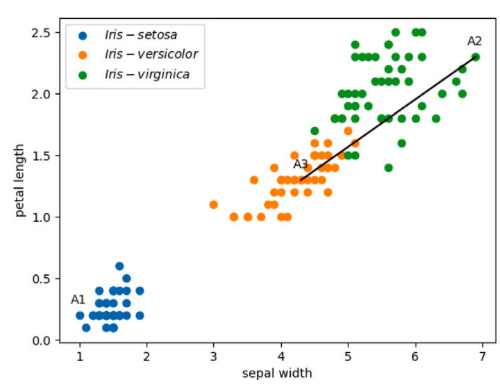

Fig. 3. Search reasonable initial prototypes $\{A1, A2, A3\}$.

q_{lj_c} is an intuitionistic fuzzy set that includes the j_c categorical attribute values, membership and non-membership degree of the attribute values in cluster C_l. q_{lj_c} reflects the distribution of the j_c categorical attribute in cluster C_l.

The intuitionistic distribution centroid reflects the aggregation of different attribute values in the cluster and the degree of dispersion of those values. It contains more information than the traditional distribution centroid.

The proof that q_{lj_c} is an intuitionistic fuzzy set is as follows:

Property 1. *Let the categorical attributes value distribution be* $q_{lj_c} \in Q_{l_c}$*, for all* $\left\langle a^k_{j_c}, \mu^k_{lj_c}, v^k_{lj_c}\right\rangle \in q_{lj_c}$*, we have* $0 \leq \mu^k_{lj_c} + v^k_{lj_c} \leq 1$.

Table 1
Artificial categorical data set.

$cluster_1$			$cluster_2$		
$Attribute_1$	$Attribute_2$	$Attribute_3$	$Attribute_1$	$Attribute_2$	$Attribute_3$
a	d	g	a	d	g
a	d	g	a	d	g
a	d	h	a	e	g
a	e	h	a	f	g
a	e	h	a	f	g
b	e	h	c	f	h
b	e	g	c	f	h
b	e	h	c	f	h
b	e	h	c	f	h
b	e	g	c	f	g

Proof. By Definition 1, we have $\mu^k_{lj_c} = \left|\left[a^k_{j_c}\right] \cap C_l\right|^2 / \left|\left[a^k_{j_c}\right]\right| |C_l|$ and $\upsilon^k_{lj_c} = \sum_{s\neq l}\left(\left|\left[a^k_{j_c}\right] \cap C_s\right| / \left|\left[a^k_{j_c}\right]\right|\right)^2$. Obviously, $\mu^k_{lj_c} + \upsilon^k_{lj_c} \geq 0$.

We obviously have $\sum_{l=1}^k \left(\left|\left[a^k_{j_c}\right] \cap C_l\right| / \left|\left[a^k_{j_c}\right]\right|\right) = 1$. Since $\left|\left[a^k_{j_c}\right] \cap C_l\right| / \left|\left[a^k_{j_c}\right]\right|, \left|\left[a^k_{j_c}\right] \cap C_l\right| / |C_l| \in [0,1]$. We have $\mu^k_{lj_c} \leq \left|\left[a^k_{j_c}\right] \cap C_l\right| / \left|\left[a^k_{j_c}\right]\right|$ and $\upsilon^k_{lj_c} \leq \left|\left[a^k_{j_c}\right] \cap C_l\right| / \left|\left[a^k_{j_c}\right]\right|$. Then, $\mu^k_{lj_c} + \upsilon^k_{lj_c} \leq \sum_{l=1}^k \left(\left|\left[a^k_{j_c}\right] \cap C_l\right| / \left|\left[a^k_{j_c}\right]\right|\right)$. Hence, $\mu^k_{lj_c} + \upsilon^k_{lj_c} \leq 1$. The property of the above is established. □

We can take q_{lj_c} as an intuitionistic fuzzy set with the above.

Example 1. Table 1 shows an artificial data set with 20 objects divided into two clusters. Each object has three categorical attributes ($Attribute_1$, $Attribute_2$, and $Attribute_3$). Table 2 shows the specific form of the intuitionistic distribution centroid obtained for each cluster according to Definition 1.

Take the calculation of $Attribute_1$ in $cluster_1$ as an example to show the calculation process.

The membership of $cluster_1$: $\mu_a = \frac{5\times5}{10\times10} = \frac{1}{4}$, $\mu_b = \frac{5\times5}{5\times10} = \frac{1}{2}$, $\mu_c = \frac{0\times0}{5\times10} = 0$.

The non-membership of $cluster_2$: $\upsilon_a = \left(\frac{5}{10}\right)^2 = \frac{1}{4}$, $\upsilon_b = \left(\frac{0}{5}\right)^2 = 0$, $\upsilon_c = \left(\frac{5}{5}\right)^2 = 1$.

In $cluster_1$, the membership of a and b should be the same because they have the same number. However, this is different from the global distribution.(Refer to Table 1) The value a has the same proportion in $cluster_1$ and $cluster_2$. Therefore, the distinction of a in the two clusters is less significant than b and c in Table 2. And the membership and non-membership in Table 2 conform to this idea. It is reasonable for the distribution of attribute values. The intuitionistic distribution centroid of the remaining attributes are detailed in Table 2.

4.2. Iterative weight adjustment strategy combining intra-cluster complexity and inter-cluster similarity

The following attribute weight strategy combines the information within and between clusters. Next, we explain the calculation method of attribute complexity and similarity in this paper and obtain a comprehensive measure. The larger the comprehensive measure value, the lower the attribute priority.

4.2.1. Iterative weight adjustment strategy for numerical attributes

When comparing the degree of dispersion of different attributes, it is not appropriate to use standard deviation or variance directly for comparison if dimensions are different. The main advantage of the coefficient of variation method is that it can handle multi-dimensional data and does not depend on the dimension of the data. At the same time, the coefficient of variation method can clearly reveal the relative importance of each indicator. For the calculation of numerical attribute complexity, we choose the coefficient of variation that measures the degree of dispersion to operate.

The coefficient of variation measures the dispersion of a probability distribution and is defined as the ratio of the standard deviation to the mean. In cluster C_l, the coefficient of variation $\xi_{j_n}(C_l)$ of numerical attribute $A^n_{j_n}$ is the ratio of the standard deviation $\sigma_{j_n}(C_l)$ to the mean $u_{j_n}(C_l)$. The specific form is as follows:

$$\xi_{j_n}(C_l) = \begin{cases} 0 & u_{j_n}(C_l) = 0 \\ \frac{\sigma_{j_n}(C_l)}{u_{j_n}(C_l)} & otherwise \end{cases} \tag{5}$$

where, $\sigma^2_{j_n}(C_l) = \frac{1}{|C_l|}\sum_{x_i\in C_l}\left(x_{ij_n} - u_{j_n}(C_l)\right)^2$, $u_{j_n}(C_l) = \frac{1}{|C_l|}\sum_{x_i\in C_l} x_{ij_n}$.

For the convenience of calculation, we use the normalized form. The coefficient of variation $\xi_{j_n}(C_l)$ can be normalized as $\xi^*_{j_n}(C_l) = \frac{\xi_{j_n}(C_l)}{\xi_{j_n \max}}$, and $\xi_{j_n \max} = \max_{1\leq i\leq k} \xi_{j_n}(C_i)$. The normalized variation coefficient and the proportion of clusters are used to construct the numerical attribute complexity.

Definition 2 (*Intra-Cluster Complexity of Numerical Attributes [12]*). The normalized coefficient of variation of the numerical attribute $A^n_{j_n}$ in cluster C_l is $\xi^*_{j_n}(C_l)$, then the intra-cluster complexity measure of $A^n_{j_n}$ is $\xi^n\left(A^n_{j_n}\right)$, and its specific form is as follows:

$$\xi^n\left(A^n_{j_n}\right) = \sum_{i=1}^k \alpha_i\left(\xi^*_{j_n}(C_i)\right) \in [0,1] \tag{6}$$

where, $\alpha_i = \frac{|C_i|}{n}$, $|C_i|$ is the number of elements in the C_i and n is the number of all objects.

Example 2. We normalize the attributes of the Iris data set and calculate the coefficient of variation. Detailed demonstration of the normalization of the coefficient of variation is provided later. The corresponding classes are in Table 3 is as follows: C_1: Iris-setosa class, C_2: Iris-versicolor class, C_3: Iris-virginica class.

When calculating the normalized coefficient of variation, the numerator selects the value of the corresponding cluster, and the denominator is the largest value in the corresponding row.

The normalized coefficient of variation of C_1:

sepal length: $\frac{0.494}{0.494} = 1$. sepal width: $\frac{0.266}{0.403} \approx 0.660$. petal length: $\frac{0.370}{0.370} = 1$. petal width: $\frac{0.737}{0.737} = 1$.

Repeat the above steps, and we will get the Table 4.

We construct a global-local similarity formula to calculate inter-cluster numerical attribute similarity. Its specific description is as follows. We know there may be noise objects in a cluster, which will have an opposite effect on calculating attribute similarity between clusters. Inspired by the idea of granular-ball [25,26], the radius of the granular-ball is the average distance from the objects in the cluster to the center. We apply the idea to the numerical attributes of each cluster.

As for the attribute A_{j_n} in the cluster C_l, we calculate the global and local cluster attribute radius:

$$r^{global}_{j_n}(C_l) = mean\left\{d_{j_n}\left(x_i, x_{C_l}\right) | x_i \in C_l\right\} \tag{7}$$

$$r^{local}_{j_n}(C_l) = mean\left\{d_{j_n}\left(x_i, x_{C_l}\right) \middle| x_i \in C_l \wedge d_{j_n}\left(x_i, x_{C_l}\right) \leq r^{global}_{j_n}(C_l)\right\} \tag{8}$$

According to the global radius, we can find objects that better represent the numerical attributes of cluster. We think these objects are "good".

Table 2
Intuitionistic distribution centroids.

	$cluster_1$	$cluster_2$
$Attribute_1$	$\left\{\left\langle a,\frac{1}{4},\frac{1}{4}\right\rangle,\left\langle b,\frac{1}{2},0\right\rangle,\langle c,0,1\rangle\right\}$	$\left\{\left\langle a,\frac{1}{4},\frac{1}{4}\right\rangle,\langle b,0,1\rangle,\left\langle c,\frac{1}{2},0\right\rangle\right\}$
$Attribute_2$	$\left\{\left\langle d,\frac{9}{50},\frac{1}{25}\right\rangle,\left\langle e,\frac{49}{80},\frac{1}{100}\right\rangle,\langle f,0,1\rangle\right\}$	$\left\{\left\langle d,\frac{2}{25},\frac{9}{100}\right\rangle,\left\langle e,\frac{1}{80},\frac{49}{100}\right\rangle,\left\langle f,\frac{7}{10},0\right\rangle\right\}$
$Attribute_3$	$\left\{\left\langle g,\frac{4}{25},\frac{9}{25}\right\rangle,\left\langle h,\frac{9}{25},\frac{4}{25}\right\rangle\right\}$	$\left\{\left\langle g,\frac{9}{25},\frac{4}{25}\right\rangle,\left\langle h,\frac{4}{25},\frac{9}{25}\right\rangle\right\}$

Table 3
The coefficient of variation of numerical attributes in Iris data set.

Attribute	$\xi(C_l)$	$\xi(C_2)$	$\xi(C_3)$
Sepal length	0.494	0.312	0.275
Sepal width	0.266	0.403	0.328
Petal length	0.370	0.143	0.120
Petal width	0.737	0.160	0.141

Table 4
The normalized coefficient of variation of numerical attributes in Iris data set.

Attribute	$\xi^*(C_l)$	$\xi^*(C_2)$	$\xi^*(C_3)$
Sepal length	1.000	0.632	0.557
Sepal width	0.660	1.000	0.814
Petal length	1.000	0.386	0.324
Petal width	1.000	0.217	0.191

For any two clusters C_{l_1}, C_{l_2}, we give the global and local cluster attribute distance:

$$d_{j_n}^{global}\left(C_{l_1}, C_{l_2}\right) = mean\left\{ d_{j_n}\left(x_i, x_{C_{l_2}}\right) \middle| x_i \in C_{l_1} \right\} \tag{9}$$

$$d_{j_n}^{local}\left(C_{l_1}, C_{l_2}\right) = mean\left\{ d_{j_n}\left(x_i, x_{C_{l_2}}\right) \middle| x_i \in C_{l_1} \wedge d_{j_n}\left(x_i, x_{C_{l_1}}\right) \le r_{j_n}^{global}\left(C_{l_1}\right) \right\} \tag{10}$$

Since the global-local distance is asymmetric, the corresponding $d_{j_n}^{global}\left(C_{l_2}, C_{l_1}\right)$ and $d_{j_n}^{local}\left(C_{l_2}, C_{l_1}\right)$ are similar to the above calculation.

In the following construction, we tend to increase the influence of “good” objects, that is, to add local influence while considering the global one. First, we use the $\frac{r_{j_n}^{global}\left(C_{l_1}\right)}{d_{j_n}^{global}\left(C_{l_2}, C_{l_1}\right)} + \frac{r_{j_n}^{global}\left(C_{l_2}\right)}{d_{j_n}^{global}\left(C_{l_1}, C_{l_2}\right)}$ as an indicator of global similarity. For the first part of this formula, it is evident that the distance from the attribute value distribution of C_{l_1} to the center is divided by the distance from C_{l_2} to the center of C_{l_1}. The same is true for the second part. The larger the ratio is, the more similar the two clusters are. Next, improving the impact of the “good” part improves the noise resistance performance. Next, we multiply the local and global effects to construct the following formula.

$$h_{j_n}\left(C_{l_1}, C_{l_2}\right) = \frac{r_{j_n}^{global}\left(C_{l_1}\right) r_{j_n}^{local}\left(C_{l_1}\right)}{d_{j_n}^{global}\left(C_{l_2}, C_{l_1}\right) d_{j_n}^{local}\left(C_{l_2}, C_{l_1}\right)} + \frac{r_{j_n}^{global}\left(C_{l_2}\right) r_{j_n}^{local}\left(C_{l_2}\right)}{d_{j_n}^{global}\left(C_{l_1}, C_{l_2}\right) d_{j_n}^{local}\left(C_{l_1}, C_{l_2}\right)} \tag{11}$$

We construct the following similarity through $h_{j_n}\left(C_{l_1}, C_{l_2}\right)$ for consistency with the complexity of numerical attributes.

$$S_{j_n}\left(C_{l_1}, C_{l_2}\right) = \begin{cases} 1 - e^{-h_{j_n}\left(C_{l_1}, C_{l_2}\right)} & d_{j_n}^* \ne 0 \\ 1 & d_{j_n}^* = 0 \end{cases} \tag{12}$$

where, $d_{j_n}^* = d_{j_n}^{local}\left(C_{l_1}, C_{l_2}\right) d_{j_n}^{local}\left(C_{l_2}, C_{l_1}\right)$. The following definition is given by applying $S_{j_n}\left(C_{l_1}, C_{l_2}\right)$ and the proportion of clusters.

Table 5
Similarity between different clusters of four attributes in Iris data set.

Attribute	$S(C_1, C_2)$	$S(C_1, C_3)$	$S(C_2, C_3)$
Sepal length	0.125	0.056	0.368
Sepal width	0.154	0.253	0.690
Petal length	0.011	0.005	0.087
Petal width	0.011	0.011	0.080

For the convenience of the narrative definition, we define a set D:

$$D = \left\{ d_{ij} \middle| d_{ij} = \left(C_i, C_j\right), 1 \le i \le k-1, i+1 \le j \le k \right\} \tag{13}$$

where C_i, C_j are clusters in the set $\{C_1, C_2, \ldots, C_k\}$, $\left(C_i, C_j\right)$ is the combined form of the different clusters.

Definition 3 (*Inter-Cluster Similarity of Numerical Attributes*)**.** Assume the set of clusters is$\{C_1, C_2, \ldots, C_k\}$. The inter-cluster similarity of the numerical attribute $A_{j_n}^n$ can be defined with the above similarity. Related formulas can be found below:

$$S^n\left(A_{j_n}^n\right) = \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} \theta_{ij} S_{j_n}\left(d_{ij}\right) \in [0,1] \tag{14}$$

where, k is the number of clusters, $d_{ij} = \left(C_i, C_j\right)$, $\theta_{ij} = \frac{\left|d_{ij}\right|}{D_{sum}}$, θ_{ij} is the weight in Eq. (14). According to the set of D: $\left|d_{ij}\right| = \left|C_i\right| + \left|C_j\right|$, $D_{sum} = \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} \left|d_{ij}\right|$, where $\left|C_i\right|, \left|C_j\right|$ is the number of objects in its corresponding cluster.

Example 3. To have consistency in attribute comparison, we normalized the attributes of the Iris data set and showed them graphically. Fig. 4 shows the distribution of the four attributes of the true label. The red star represents the center value, and the red line represents the global radius. By comparing the following figure, we can find that the denser the distribution of attribute values in the cluster, the smaller its radius.

The corresponding classes are in Table 5 is as follows: C_1: Iris-setosa class, C_2: Iris-versicolor class, C_3: Iris-virginica class. The similarity of the four attributes is calculated in Table 5 according to Eq. (14). The similarity of sepal length is 0.283, sepal width is 0.629, petal length is 0.061, petal width is 0.061.

According to the table and figure representation of the Iris data set, we can find that the similarity between the clusters in the third and fourth attributes is slight, and the corresponding attribute value distribution is dense. So it can better distinguish between the three classes. It should account for more in the weight allocation later. The similarity between the two and three classes is remarkable in the corresponding second attribute. The ability to distinguish the three clusters is weak, so the assigned weight is negligible.

$\xi^n\left(A_{j_n}^n\right)$ is constructed from the coefficient of variation. The more chaotic the attribute value distribution is, the larger the $\xi^n\left(A_{j_n}^n\right)$ value is. $S^n\left(A_{j_n}^n\right)$ is obtained by integrating the inter-cluster similarity we defined. For different clusters, the more similar the attribute value distribution is, the larger the $S^n\left(A_{j_n}^n\right)$ value is. We know that attribute

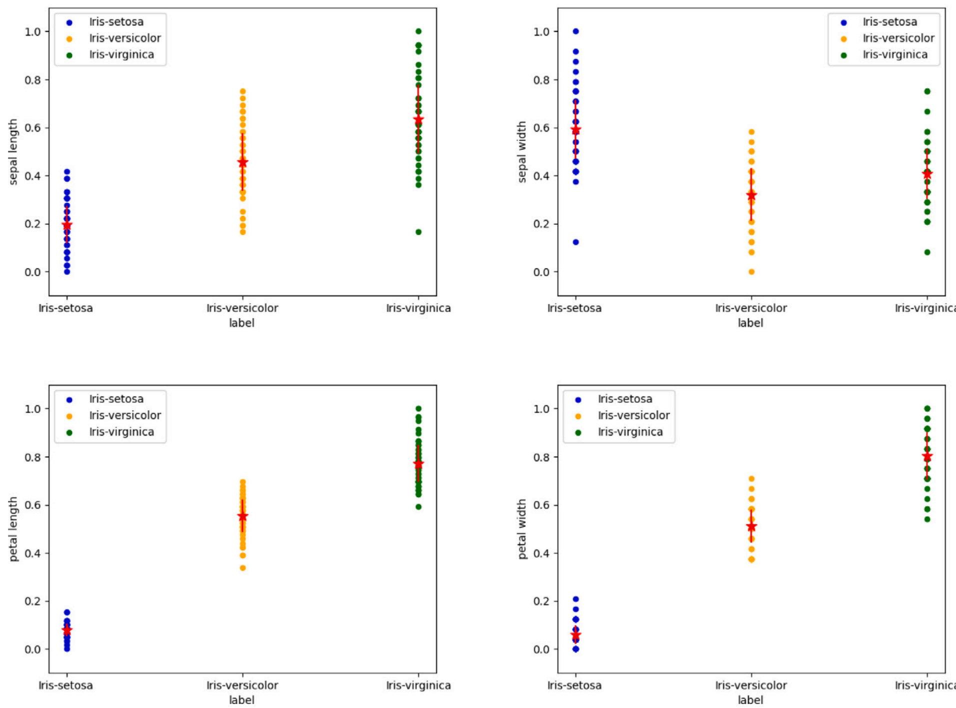

Fig. 4. Four attributes radius of the true label in the Iris data set. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

values are not concentrated, and attributes that are less clear in distinguishing different clusters are not friendly to clustering tasks, so we combine these two formulas and refer to the method of Huang [22]. The weights of the attributes are negatively correlated with our composite measure.

Definition 4 (*Numerical Attributes Weight*)**.** Assume $\varphi^n\left(A^n_{j_n}\right)$ is a comprehensive measure of the union of $\xi^n\left(A^n_{j_n}\right)$ and $S^n\left(A^n_{j_n}\right)$ through $\gamma \in [0,1]$. Numerical attribute weight num_{j_n} should increase with the decrease of $\varphi^n\left(A^n_{j_n}\right)$. The related formulas are shown in the following:

$$\varphi^n\left(A^n_{j_n}\right) = \gamma\xi^n\left(A^n_{j_n}\right) + (1-\gamma)\,S^n\left(A^n_{j_n}\right) \in [0,1] \tag{15}$$

$$num_{j_n} = \begin{cases} 0 & \varphi^n\left(A^n_{j_n}\right) = 0 \\ \dfrac{1}{\sum_{t_n=1}^{num\_n}\left[\varphi^n\left(A^n_{j_n}\right)/\varphi^n\left(A^n_{t_n}\right)\right]^{1/\beta-1}} & otherwise \end{cases} \tag{16}$$

where, num_n is the number of $\varphi^n\left(A^n_{t_n}\right) \neq 0$, and $\sum_{t_n=1}^{num\_n} num_{t_n} = 1$.

Assume $\left\{num_{1_n}, num_{2_n}, \dots, num_{m_n}\right\}$ is the weight of the corresponding m_n numerical attributes. Consider the number of numerical attributes corresponding weight should be $num^*_{i_n} = \left(num_{i_n}\right) \times m_n$. Get the new weight $\left\{num^*_{1_n}, num^*_{2_n}, \dots, num^*_{m_n}\right\}$.

4.2.2. Iterative weight adjustment strategy for categorical attributes

Similarly, we calculate the complexity and the similarity of categorical attributes to assign weights. When calculating the complexity of categorical attributes, we combine the intuitionistic distribution centroids proposed in this paper with the entropy measure of intuitionistic fuzzy sets [27]. We assume the intuitionistic distribution centroid of cluster C_l is $Q_{l_c} = \left\{q_{l1_c}, q_{l2_c}, \dots, q_{lm_c}\right\}$, then the intuitionistic fuzzy entropy for q_{lj_c} can be defined with formulas:

$$Entropy\left(q_{lj_c}\right) = \frac{\sum_{i=1}^{t}\left(\mu^k_{lj_c} \wedge \upsilon^k_{lj_c}\right)}{\sum_{i=1}^{t}\left(\mu^k_{lj_c} \vee \upsilon^k_{lj_c}\right)} \tag{17}$$

where, $q_{lj_c} = \left\{\left\langle a^1_{j_c}, \mu^1_{l_ij_c}, \upsilon^1_{l_ij_c}\right\rangle, \left\langle a^2_{j_c}, \mu^2_{l_ij_c}, \upsilon^2_{l_ij_c}\right\rangle, \dots, \left\langle a^t_{j_c}, \mu^t_{l_ij_c}, \upsilon^t_{l_ij_c}\right\rangle\right\}$.

Intuitionistic fuzzy entropy describes the distribution of categorical attribute values. We can find that the larger the entropy value, the more unfocused the distribution of categorical attribute values across the clusters. We apply the entropy measure to construct the following complexity.

Definition 5 (*Intra-Cluster Complexity of Categorical Attributes*)**.** The intuitionistic fuzzy entropy of the categorical attribute $A^c_{j_c}$ in $Q_{l_c} = \left\{q_{l1_c}, q_{l2_c}, \dots, q_{lm_c}\right\}$ is $Entropy\left(q_{lj_c}\right)$, then the intra-cluster complexity measure of the j_c attribute $A^c_{j_c}$ is $E^c\left(A^c_{j_c}\right)$, and its specific form is as follows:

$$E^c\left(A^c_{j_c}\right) = \sum_{i=1}^{k}\alpha_i\left(Entropy\left(q_{ij_c}\right)\right) \in [0,1] \tag{18}$$

where, $\alpha_i = \frac{|C_i|}{n}$, $a^k_{j_c}$, $|C_i|$ is the number of elements in the C_l and n is the number of all objects.

We calculate the similarity between clusters of categorical attributes based on the similarity of intuitionistic fuzzy sets in [28]. We assume $q_{l_1j_c}$ and $q_{l_2j_c}$ are intuitionistic fuzzy sets of the categorical attribute $A^c_{j_c}$ in the cluster C_{l_1} and C_{l_2}. The similarity of $q_{l_1j_c}$ and $q_{l_2j_c}$ is given below:

$$S_{j_c}\left(C_{l_1}, C_{l_2}\right) = 1 - \frac{\sum_{i=1}^{t}\left|\mu^i_{l_1j_c} - \upsilon^i_{l_1j_c} - \left(\mu^i_{l_2j_c} - \upsilon^i_{l_2j_c}\right)\right|}{2t} \tag{19}$$

where, $q_{l_ij_c} = \left\{\left\langle a^1_{j_c}, \mu^1_{l_ij_c}, \upsilon^1_{l_ij_c}\right\rangle, \left\langle a^2_{j_c}, \mu^2_{l_ij_c}, \upsilon^2_{l_ij_c}\right\rangle, \dots, \left\langle a^t_{j_c}, \mu^t_{l_ij_c}, \upsilon^t_{l_ij_c}\right\rangle\right\}$ $(1 \le i \le 2)$. Like Definition 3, attribute similarity is defined below using the abovementioned formula.

Definition 6 (*Inter-Cluster Similarity of Categorical Attributes*)**.** Assume the set of clusters is$\left\{C_1, C_2, \dots, C_k\right\}$. The inter-cluster similarity of the categorical attribute $A^c_{j_c}$ can be defined with the similarity of intuitionistic fuzzy sets. Related formulas can be found below:

$$S^c\left(A^c_{j_c}\right)=\sum_{i=1}^{k-1}\sum_{j=i+1}^{k}\theta_{ij}S_{j_c}\left(d_{ij}\right)\in[0,1] \tag{20}$$

where, k is the number of clusters, $d_{ij}=\left(C_i,C_j\right)$, $\theta_{ij}=\frac{\left|d_{ij}\right|}{D_{sum}}$, θ_{ij} is the weight in Eq. (20). According to the set of D: $\left|d_{ij}\right|=\left|C_i\right|+\left|C_j\right|$, $D_{sum}=\sum_{i=1}^{k-1}\sum_{j=i+1}^{k}\left|d_{ij}\right|$, where $\left|C_i\right|,\left|C_j\right|$ is the number of objects in its corresponding cluster.

The higher the similarity and complexity, the worse the expression of distinction. Hence, weight is negatively correlated with complexity and similarity.

Definition 7 (*Categorical Attributes Weight*). Assume $\varphi^c\left(A^c_{j_c}\right)$ is a comprehensive measure of the union of $E^c\left(A^c_{j_c}\right)$ and $S^c\left(A^c_{j_c}\right)$ through $\gamma\in[0,1]$. Categorical attribute weight cat_{j_c} should increase with the decrease of $\varphi^c\left(A^c_{j_c}\right)$. The related formulas are shown in the following:

$$\varphi^c\left(A^c_{j_c}\right)=\gamma E^c\left(A^c_{j_c}\right)+(1-\gamma)S^c\left(A^c_{j_c}\right)\in[0,1] \tag{21}$$

$$cat_{j_c}=\begin{cases}0 & \varphi^c\left(A^c_{j_c}\right)=0\\ \dfrac{1}{\sum_{t_c=1}^{cat\_n}\left[\varphi^c\left(A^c_{j_c}\right)/\varphi^c\left(A^c_{t_c}\right)\right]^{1/\beta-1}} & otherwise\end{cases} \tag{22}$$

where, cat_n is the number of $\varphi^c\left(A^c_{t_c}\right)\neq 0$, and $\sum_{t_c=1}^{cat\_n}cat_{t_c}=1$.

Let $\left\{cat_{1_c},cat_{2_c},\dots,cat_{m_c}\right\}$ be weights of the corresponding m_c categorical attributes. Consider the number of categorical attributes corresponding weight should be $cat^*_{i_c}=\left(cat_{i_n}\right)\times m_c$. Get the new weight $\left\{cat^*_{1_c},cat^*_{2_c},\dots,cat^*_{m_c}\right\}$.

$$S=\left\{num^*_{1_n},num^*_{2_n},\dots,num^*_{m_n},cat^*_{1_c},cat^*_{2_c},\dots,cat^*_{m_c}\right\} \tag{23}$$

We get the attribute weights by combining the new weights for numerical and categorical attributes. For the numerical attributes, we refer to the idea of [12] for the complexity part, while for the similarity part, compared with the traditional construction method [10,22]. We pay more attention to the influence brought by local representative objects. The weights of numerical attributes constructed in this paper are more anti-noise. The idea of complexity and similarity of categorical attributes is constructed by intuitionistic fuzzy measurement, which is more informative than the traditional distribution centroid [11,12]. The constructing weights show excellent results on the experimental part of the pure numerical and categorical attribute data set. We can argue that this idea is more suitable than the traditional construction of attributes.

Example 4. An artificial data set is shown in Table 1. The specific form of the intuitionistic distribution centroid is shown in Table 2.

$Attribute_1$ shown in Table 1, b and c are distinguishable in terms of their importance in the two clusters, but for a, it is impossible to distinguish the difference in the two clusters. Therefore, the complexity and similarity of $Attribute_1$ are relatively high in Tables 6 and 7.

About $Attribute_2$, e prefers $cluster_1$ and f prefers $cluster_2$. Although d is not very different in the two clusters, it can also be found to favor $cluster_1$.$Attribute_2$ has the least similarity.

The difference between g and h in $Attribute_3$ is small, and it is impossible to clearly distinguish the difference between the two attribute values, so $Attribute_3$ is less meaningful for clustering. So, its similarity is the largest.

The corresponding φ^c and weights are obtained according to Definition 7. The new weights are obtained by multiplying the weights by the corresponding number of categorical attributes. Their contents are in Table 8.

4.3. Categorical attribute distance

Definition 8 (*Membership and Non-Membership Distance*). Let x_i be an object in the data set X and the value domain of the categorical attribute $A^c_{j_c}$ is $Dom\left(A^c_{j_c}\right)=\left\{a^1_{j_c},a^2_{j_c},\dots,a^t_{j_c}\right\}$. The $A^c_{j_c}$ value of object x_i be x_{ij_c}, then the membership distance $d_1\left(x_{ij_c},q_{lj_c}\right)$ and non-membership distance $d_2\left(x_{ij_c},q_{lj_c}\right)$ of $A^c_{j_c}$ about object x_i and intuitionistic distribution centroid Q_{l_c} are as follows:

$$d_1\left(x_{ij_c},q_{lj_c}\right)=\sum_{k=1}^{t}\theta_1\left(x_{ij_c},a^k_{j_c}\right)\quad d_2\left(x_{ij_c},q_{lj_c}\right)=\sum_{k=1}^{t}\theta_2\left(x_{ij_c},a^k_{j_c}\right) \tag{24}$$

$$\theta_1\left(x_{ij_c},a^k_{j_c}\right)=\begin{cases}0 & x_{ij_c}=a^k_{j_c}\\ \mu^k_{lj_c} & x_{ij_c}\neq a^k_{j_c}\end{cases}\quad \theta_2\left(x_{ij_c},a^k_{j_c}\right)=\begin{cases}\upsilon^k_{lj_c} & x_{ij_c}=a^k_{j_c}\\ 0 & x_{ij_c}\neq a^k_{j_c}\end{cases} \tag{25}$$

where, $q_{lj_c}=\left\{\left\langle a^1_{j_c},\mu^1_{l_ij_c},\upsilon^1_{l_ij_c}\right\rangle,\left\langle a^2_{j_c},\mu^2_{l_ij_c},\upsilon^2_{l_ij_c}\right\rangle,\dots,\left\langle a^t_{j_c},\mu^t_{l_ij_c},\upsilon^t_{l_ij_c}\right\rangle\right\}$.

Definition 9 (*Composite Distance*). Let x_i be an object in the data set X and the value domain of the categorical attribute $A^c_{j_c}$ is $Dom\left(A^c_{j_c}\right)=\left\{a^1_{j_c},a^2_{j_c},\dots,a^t_{j_c}\right\}$. The $A^c_{j_c}$ value of object x_i be x_{ij_c}, then the composite distance $d\left(x_{ij_c},q_{lj_c}\right)$ of $A^c_{j_c}$ about object x_i and intuitionistic distribution centroid Q_{l_c} is as follows:

$$d\left(x_{ij_c},q_{lj_c}\right)=\mu d_1\left(x_{ij_c},q_{lj_c}\right)+(1-\mu)d_2\left(x_{ij_c},q_{lj_c}\right) \tag{26}$$

where, $0\le\mu\le1$, $q_{lj_c}=\left\{\left\langle a^1_{j_c},\mu^1_{l_ij_c},\upsilon^1_{l_ij_c}\right\rangle,\left\langle a^2_{j_c},\mu^2_{l_ij_c},\upsilon^2_{l_ij_c}\right\rangle,\dots,\left\langle a^t_{j_c},\mu^t_{l_ij_c},\upsilon^t_{l_ij_c}\right\rangle\right\}$.

Example 5. An artificial data set is shown in Table 1. The specific form of the intuitionistic distribution centroid is shown in Table 2.

The distances from $x_0=(b,e,h)$ to two clusters are shown in Table 9 as an example to illustrate the defined distances. The final weight distance can be obtained by combining the weights in Table 8.

The distance from x_0 to two clusters is as follows:

From x_0 to $clusters_1$: $1.1274\times0.125+1.269\times0.095+0.5826\times0.160=0.354696$.

From x_0 to $clusters_2$: $1.1274\times0.875+1.269\times0.635+0.5826\times0.360=2.002026$.

In the clustering process, x_0 will be divided into $clusters_1$.

4.4. Intuitive-K-prototypes algorithm

The cost function and distance measure for the algorithm are defined in Eqs. (27)–(29)

$$Cost(U,Q,S)=\sum_{l=1}^{k}\sum_{i=1}^{n}u_{il}\Phi\left(x_i,Q_l\right) \tag{27}$$

$$\Phi\left(x_i,Q_l\right)=\Phi^n\left(x_i,Q_l\right)+\Phi^c\left(x_i,Q_l\right) \tag{28}$$

$$\Phi^n\left(x_i,Q_l\right)=\sum_{j_n=1_n}^{m_n}num^*_{j_n}\left|x_{ij_n}-q_{ij_n}\right|\quad \Phi^c\left(x_i,Q_l\right)=\sum_{j_c=1_c}^{m_c}cat^*_{j_c}d\left(x_{ij_c},q_{lj_c}\right) \tag{29}$$

The membership matrix $U=\left\{u_{il}\right\}_{n\times k}$ that represents the membership of x_i and C_l, and the membership is $u_{il}=\begin{cases}1 & x_i\in C_l\\ 0 & x_i\notin C_l\end{cases}$. The prototypes in Q consist of the mean value of the numerical attribute $Q_n=\left\{u\left(C_l\right)\right\}_{1\times k}$ and the intuitive distribution centroid $Q_c=\left\{Q_{l_c}\right\}_{1\times k}$, where, $u\left(C_l\right)=\left\{u_{j_n}\left(C_l\right)\right\}_{1\times m_n}$. The attribute weights S are defined by Eq. (23). The specific flow of the proposed algorithm is as follows:

Table 6
Intra-cluster complexity of categorical attributes.

	$cluster_1$	$cluster_2$	$complexity$
$Attribute_1$	$\frac{(1/4)+0+0}{(1/4)+(1/2)+1} = \frac{1}{7}$	$\frac{(1/4)+0+0}{(1/4)+(1/2)+1} = \frac{1}{7}$	$\left(\frac{1}{2}\right)\left(\frac{1}{7}\right)+\left(\frac{1}{2}\right)\left(\frac{1}{7}\right) \approx 0.143$
$Attribute_2$	$\frac{(1/25)+(1/100)+0}{(9/50)+(49/80)+1} = \frac{20}{717}$	$\frac{(2/25)+(1/80)+0}{(9/100)+(49/100)+(7/10)} = \frac{37}{512}$	$\left(\frac{1}{2}\right)\left(\frac{20}{717}\right)+\left(\frac{1}{2}\right)\left(\frac{37}{512}\right) \approx 0.050$
$Attribute_3$	$\frac{(4/25)+(4/25)}{(9/25)+(9/25)} = \frac{4}{9}$	$\frac{(4/25)+(4/25)}{(9/25)+(9/25)} = \frac{4}{9}$	$\left(\frac{1}{2}\right)\left(\frac{4}{9}\right)+\left(\frac{1}{2}\right)\left(\frac{4}{9}\right) \approx 0.444$

Table 7
Inter-cluster similarity of categorical attributes.

	$similarity$
$Attribute_1$	$\left(1-\frac{0+(3/2)+(3/2)}{6}\right) = 0.500$
$Attribute_2$	$\left(1-\frac{(3/20)+(27/25)+(17/10)}{6}\right) \approx 0.512$
$Attribute_3$	$\left(1-\frac{(2/5)+(2/5)}{4}\right) = 0.800$

Table 8
Categorical attributes weight.

$\beta = 2 \quad \gamma = 0.5$	φ^c	$weight$	$new\ weight$
$Attribute_1$	0.3215	0.3758	1.1274
$Attribute_2$	0.281	0.4230	1.269
$Attribute_3$	0.622	0.1942	0.5826

Table 9
Different distance.

	$cluster_1$	$cluster_2$
x_0	$0.5d_1 + 0.5d_2$	$0.5d_1 + 0.5d_2$
$Attribute_1$	$\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)+0 = 0.125$	$\left(\frac{1}{2}\right)\left(\frac{3}{4}\right)+\left(\frac{1}{2}\right) = 0.875$
$Attribute_2$	$\left(\frac{1}{2}\right)\left(\frac{9}{50}\right)+\left(\frac{1}{2}\right)\left(\frac{1}{100}\right) = 0.095$	$\left(\frac{1}{2}\right)\left(\frac{78}{100}\right)+\left(\frac{1}{2}\right)\left(\frac{49}{100}\right) = 0.635$
$Attribute_3$	$\left(\frac{1}{2}\right)\left(\frac{4}{25}\right)+\left(\frac{1}{2}\right)\left(\frac{4}{25}\right) = 0.160$	$\left(\frac{1}{2}\right)\left(\frac{9}{25}\right)+\left(\frac{1}{2}\right)\left(\frac{9}{25}\right) = 0.360$

Step 1: Randomly select the initial prototype in $X = \{x_1, x_2, \dots, x_n\}$.

Step 2: The objects are divided into different parts and get initial clusters by Eq. (1).

Step 3: Update intuitionistic distribution centroids Q according to the clustering results and update attribute weights by Eqs. (5)–(23).

Step 4: The distance between data objects and different distribution centroids is recalculated by Eqs. (28)–(29), and the objects are assigned to the closest cluster to obtain a new clustering result.

Step 5: Check whether the clustering results are consistent. If the results are the same, the clustering results will be output. If the results are different, repeat Steps 3 and 4 until the results are consistent or the maximum number of iterations is reached.

4.5. Algorithm time complexity analysis

For the clustering algorithm 2, time complexity analysis mainly includes updating the intuitionistic distribution centroid and membership matrix in each iteration. The cost of updating the centroid and partition matrix are $O(kmn)$ and $O\left(2k\left(m_n + \left(Nm_c\right)\right)n\right)$. The overall time complexity is $O\left(\left(mn + 2m_n + 2Nm_c\right)lkn\right)$. The k is the number of clusters; m_n is the number of numerical attributes; m_c is the number of categorical attributes; m is the number of all attributes; $N = \max(t)$ is the maximum value number of all categorical attributes. n is the number of data objects. l is the number of iterations required for converging our algorithm. So, the time complexity is linear $O(n)$.

Algorithm 2 Intuitive-K-prototypes

Input: mixed data sets $X = \{x_1, x_2, \dots, x_n\}$, number of clusters k, distance parameter μ, balancing parameter γ,weight index β , maximum number of iterations l.

Output: cluster label $y = \{y_1, y_2, \dots, y_n\}$.

Initial: membership matrix $U = \{0\}_{n\times k}$, randomly selected initial prototypes set $Q = \{q^1, q^2, \dots, q^k\}$, attribute weights $S = \{1\}_{1\times m}$, number of iteration $s = 0$.

1: **for** $i = 1$ to n **do**

2: calculate the distance between x_i and q^j $(1 \le j \le k)$ by Eq. (1).

3: assign x_i to the cluster represented by the nearest prototype q^j $(1 \le j \le k)$.

4: update membership matrix U.

5: **end for**

6: $s = s + 1$.

7: update attribute weight S according to Eq. (5) – Eq. (23).

8: update Intuitionistic distribution centroids Q_c of the categorical attributes according to Eq. (2) – Eq. (4) and combine the mean of numerical attributes about clusters.

9: **for** $i = 1$ to n **do**

10: calculate the categorical attributes distance between x_i and Q_c

11: calculate the numerical attributes distance between x_i and Q_n

12: assign x_i to the nearest cluster according to the distance.

13: update membership matrix U.

14:**end for**

15:$s = s + 1$

16:**if** membership matrix U changed and $s \le l$ **then**

17: **repeat** step.7 – step.15. until membership matrix U unchanged or $s = l$.

18:**end if**

19:**return** cluster label $y = \{y_1, y_2, \dots, y_n\}$.

5. Experimental analysis

All experiments are conducted on a device equipped with an Intel(R) Core (TM) i5-8300H CPU 2.30 GHz and 16 GB RAM and simulated using Python 3.9. The intuitive-K-prototypes algorithm is evaluated using seven popular clustering validity indices on nine datasets from UCI [29].

5.1. Data sets and experimental settings

We selected seven data sets from UCI [29]. In preprocessing, missing values are filled with the mode, which also tests the noise-resistance property of clustering algorithm. IDs are removed, and attributes with only one value are also removed. Additionally, all numerical attributes have been normalized. All the data labeled “1” and “4” are removed from the Lymphography data set. The Chess dataset selects data labeled “ten” and “fifteen” as subsets.

We also compare our proposed algorithm with other traditional mixed data clustering algorithms, such as K-prototypes [7], IK-prototype [10], DPC-M [30], DP-MD-FN [31], MCFCIW [12] and SFK-prototype [15].

The K-prototypes, IK-prototypes, MCFCIW, and SFK-prototypes algorithms perform 50 experiments in pursuit of stability due to their characteristic of randomly selecting initial prototypes. Meanwhile, the

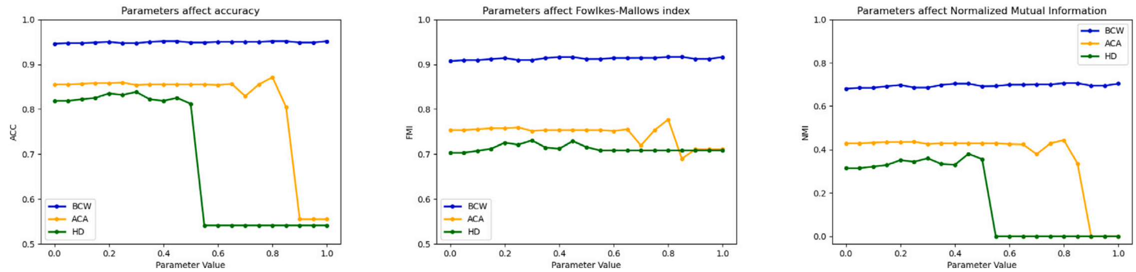

Fig. 5. The influence of distance parameters on metrics.

Table 10
Data sets from UCI.

Dataset	Clusters	Numerical attributes	Categorical attributes	Obejects
Iris	3	4	0	150
Ionosphere	2	33	0	351
BCW	2	0	9	699
Soybean	4	0	35	47
Australian Credit Approval	2	6	8	690
Chess	2	3	3	4151
Heartdisease-cleveland	2	5	8	303
Heartdisease-cleveland	5	5	8	303

DPC-M and DP-MD-FN algorithms do not need to perform multiple experiments based on their characteristics of selecting the best clustering centers (see Table 10).

The following metrics are applied for evaluation. Accuracy (AC) [32], Precision (PR) [32], Recall (RE) [32], F-score (F_β) [33], Adjusted Rand index (ARI) [34], Fowlkes–Mallows index (FMI) [35], Normalized Mutual Information (NMI) [36].

$$AC = \frac{\sum_{i=1}^{k} c_i}{n} \qquad AC \in [0,1] \tag{30}$$

where c_i is the number of clustering results that are consistent with the true label, and n is the number of data objects in the data set.

$$PR = \frac{TP}{TP+FP} \qquad PR \in [0,1] \tag{31}$$

$$RE = \frac{TP}{TP+FN} \qquad RE \in [0,1] \tag{32}$$

$$F_1 = \frac{2PR \times RE}{(PR+RE)} \qquad F_1 \in [0,1] \tag{33}$$

$$RI = \frac{TP+TN}{TP+FP+TN+FN} \qquad RI \in [0,1] \tag{34}$$

$$ARI = \frac{(RI-E(RI))}{(\max(RI)-E(RI))} \qquad ARI \in [-1,1] \tag{35}$$

$$FMI = \sqrt{\frac{TP}{TP+FN} \times \frac{TP}{TP+FP}} \qquad FMI \in [0,1] \tag{36}$$

True positives (TP) are the number of data objects correctly placed in the cluster according to their class label. True negatives (TN) are the number of data objects not put in the cluster and do not match the class label. False positives (FP) are the number of cluster instances whose class differs from the cluster, and false negatives (FN) are the number of data objects whose class is correct but are not placed in the correct cluster.

$$NMI = \frac{\sum_{r=1}^{k}\sum_{t=1}^{k} c_{r,t} \log\left(\frac{N c_{r,t}}{c_r c_t}\right)}{\sqrt{\left(\sum_{r=1}^{k} c_r \log \frac{c_r}{N}\right)}\sqrt{\left(\sum_{t=1}^{k} c_t \log \frac{c_t}{N}\right)}} \qquad NMI \in [0,1] \tag{37}$$

k represents the number of clusters, and N represents the number of data objects. $c_{r,t}$ represents the number of data objects of the same class assigned to the cluster, where c_r is the total number of data objects in the cluster r and c_t is the total number of data objects in class t.

The larger the clustering metrics above, the better the corresponding clustering performance.

5.2. The impact of distance parameter μ

The following is a brief analysis of the selection of distance parameters based on the influence of parameters on the clustering metrics of the three data sets(BCW, Australian Credit Approval, and Heart Disease-Cleveland).

In Fig. 5, the parameter has little influence on the BCW data set, indicating that the BCW data set is well distributed, which is a particular case, mainly referring to the data set of heart disease and credit card, we can find that the figure will have a cliff drop, indicating that the clustering effect will become very poor when the parameter is bigger than a certain value, referring to the value of mutual information in the third figure. The lousy result means the clustering effect is powerful and finally clustered into the same cluster. So, choosing a suitable distance parameter for this algorithm is essential. According to the figure, the parameter value should be between 0 and 0.4.

5.3. Result analysis

The following analyzes the effect of initial prototype selection on the results and compares the clustering effects between different algorithms.

5.3.1. The effect of initial prototypes selection

The effect of initial prototype selection on the clustering results: In terms of clustering performance, based on the data in Tables 11–18. Except for three data sets, the clustering performance of the proposed algorithm is improved by choosing the initial prototype. The selection of initial prototypes can even enhance the best performance of the algorithms in this paper, e.g., ACA data sets.

In Table 16 about the Heart disease data set(two clusters) data set, the selection of prototypes limits the performance of little of the algorithms. However, the metrics of both are almost the same, so we think that the clustering effect of both is the same in this case.

In Table 17 about the Chess data set, the optimal performance is the same in both cases. The poor results are due to the high percentage of unsatisfactory initial prototypes selected in the 50 process, which pulled down the metrics when averaged.

In Table 18 about Heart disease data set(five clusters) data set. We can see little difference between the metrics in this data set. The initial prototype selection can improve the AC, ARI, and FMI. As for other metrics, the maximum difference is 0.003.

We can argue that the initial prototype selection does not perform well only on the chess data set. Still, our idea can indeed improve the performance and stability.

5.3.2. Comparison of the effects between different algorithms

Regarding time complexity, the K-Prototypes, IK-Prototypes, MCF-CIW, SFK-Prototypes, and Intuitive-K-prototypes are all k-partitioned algorithms that require the iteration of centers. The time complexity of five algorithms is $O(n)$. As for DPC-M and DP-MD-FN, although they do not require the iteration of centers, which involves calculating the distance of each object from each other, so the time complexity of these

Table 11
The summary results for Iris data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Iris	K-prototypes [7]	0.898	0.887	0.887	0.892	0.716	0.811	0.742
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	IK-prototypes [10]	0.859	0.853	0.853	0.856	0.654	0.769	0.687
		(0.002)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.003)
	DPC-M [30]	0.883	0.820	0.820	0.850	0.620	0.759	0.731
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.776	0.753	0.753	0.764	0.487	0.674	0.514
	$d_c = 6\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.903	0.887	0.887	0.895	0.717	0.813	0.750
	$\gamma = 1.0$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	SFK-prototype [15]	0.897	0.889	0.889	0.893	0.720	0.813	0.737
	$Tr = 0.15$	(0.002)	(0.004)	(0.004)	(0.003)	(0.009)	(0.006)	(0.006)
	Intuitive-K-prototypes	**0.960**	**0.960**	**0.960**	**0.960**	**0.886**	**0.923**	**0.864**
	$\gamma = 0.0$	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.006)
	Intuitive-K-prototypes	**0.960**	**0.960**	**0.960**	**0.960**	**0.886**	**0.923**	**0.864**
	with select prototypes	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Table 12
The summary results for Ionosphere data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Ionosphere	K-prototypes [7]	0.701	0.717	0.711	0.709	0.176	0.605	0.133
		(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.000)	(0.002)
	IK-prototypes [10]	0.643	0.650	0.665	0.646	0.104	0.576	0.064
		(0.002)	(0.003)	(0.002)	(0.003)	(0.003)	(0.021)	(0.002)
	DPC-M [30]	0.606	0.603	0.641	0.605	0.069	0.576	0.033
	$d_c = 2\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.829	0.536	0.667	0.651	0.423	0.729	0.070
	$d_c = 4\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.654	0.662	0.673	0.658	0.116	0.581	0.075
	$\gamma = 1.0$	(0.004)	(0.005)	(0.004)	((0.004)	((0.006)	(0.003)	(0.004)
	SFK-prototype [15]	0.641	0.645	0.665	0.643	0.103	0.579	0.062
	$Tr = 0.12$	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)
	Intuitive-K-prototypes	0.712	0.677	0.710	0.692	0.169	0.631	0.129
	$\gamma = 0.0$	(0.147)	(0.097)	(0.119)	(0.122)	(0.250)	(0.132)	(0.214)
	Intuitive-K-prototypes	**0.836**	**0.730**	**0.800**	**0.780**	**0.334**	**0.740**	**0.275**
	with select prototypes	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Table 13
The summary results for Breast Cancer Wisconsin data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
BCW	K-prototypes [7]	0.939	0.909	0.931	0.924	0.739	0.887	0.635
		(0.013)	(0.025)	(0.018)	(0.019)	(0.063)	(0.025)	(0.061)
	IK-prototypes [10]	0.947	0.926	0.943	0.936	0.781	0.903	0.673
		(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)
	DPC-M [30]	0.698	0.659	0.562	0.678	−0.012	0.586	0.134
	$d_c = 1\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	–	–	–	–	–	–	–
		–	–	–	–	–	–	–
	MCFCIW [12]	**0.952**	0.934	0.949	0.943	0.802	0.912	0.696
	$\gamma = 1.0$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	SFK-prototype [15]	0.939	0.909	0.931	0.924	0.739	0.887	0.635
	$Tr = 0.15$	(0.013)	(0.025)	(0.018)	(0.019)	(0.063)	(0.025)	(0.061)
	Intuitive-K-prototypes	0.948	0.938	0.949	0.943	0.804	0.912	0.693
	$\mu = 0.2 \quad \gamma = 0.4$	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.002)	(0.004)
	Intuitive-K-prototypes	0.949	**0.939**	**0.950**	**0.944**	**0.807**	**0.914**	**0.697**
	with select prototypes	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

two algorithms is $O(n^2)$. The Intuitive-K-prototypes algorithm is ideal for time complexity.

Next, we use the experiment metrics in Tables 11–18 to compare its performance with other algorithms.

For purely numerical Iris and Ionosphere data sets, we can see Tables 11–12. For the Iris data set, the average accuracy of the Intuitive-K-prototypes algorithm reaches 0.960, with the minimum difference of 0.071 with the average accuracy of other algorithms. For the Ionosphere data set, the average accuracy of the Intuitive-K-prototypes algorithm is 0.710, which is close to the optimal average value of 0.711. However, its best accuracy is 0.829, far exceeding the best performance of other algorithms. So, the Intuitive-K-prototypes algorithm performs excellently in clustering purely numerical data sets.

For purely categorical BCW and Soybean data sets, we can see Tables 13–14. As for the DP-MD-FN algorithm, due to the defect of distance calculation, it cannot complete the clustering of the BCW data set. In the BCW data set, the accuracy of both algorithms MCFCIW and Intuitive-K-prototypes reached 0.949, and the overall difference was insignificant. We believe that both algorithms achieved the best average performance. The best accuracy of the algorithm in this paper reached 0.950 on this data set, and all metrics except PR were higher than those of other algorithms. Only the PR of MCFCIW is 0.004 higher

Table 14
The summary results for Soybean data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Soybean	K-prototypes [7]	0.968	0.971	0.967	0.970	0.930	0.947	0.950
		(0.032)	(0.029)	(0.033)	(0.030)	(0.070)	(0.053)	(0.050)
	IK-prototypes [10]	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DPC-M [30]	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
	$d_c = 6\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.958	0.971	0.957	0.964	0.875	0.906	0.910
	$\gamma = 0.5 \quad \alpha = 0.2$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	SFK-prototype [15]	0.890	0.893	0.880	0.891	0.777	0.834	0.849
	$Tr = 0.1$	(0.110)	(0.107)	(0.120)	(0.109)	(0.223)	(0.166)	(0.151)
	Intuitive-K-prototypes	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
	$\gamma = 0.5$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	Intuitive-K-prototypes	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
	with select prototypes	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Table 15
The summary results for Australian Credit Approval data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
ACA	K-prototypes [7]	0.801	0.802	0.803	0.802	0.367	0.686	0.283
	$\mu_l = 0.5$	(0.028)	(0.025)	(0.027)	(0.027)	(0.069)	(0.036)	(0.056)
	IK-prototypes [10]	0.838	0.842	0.834	0.840	0.452	0.728	0.380
		(0.023)	(0.025)	(0.027)	(0.024)	(0.072)	(0.035)	(0.054)
	DPC-M [30]	0.805	0.804	0.807	0.805	0.377	0.692	0.288
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.818	0.811	0.817	0.814	0.402	0.707	0.310
	$d_c = 2\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.802	0.803	0.802	0.803	0.377	0.692	0.309
	$\gamma = 0.7 \quad \alpha = 0.2$	(0.056)	(0.059)	(0.054)	(0.058)	(0.126)	(0.061)	(0.119)
	SFK-prototype [15]	0.809	0.806	0.809	0.807	0.381	0.695	0.297
	$Tr = 0.1$	(0.031)	(0.026)	(0.025)	(0.024)	(0.063)	(0.033)	(0.053)
	Intuitive-K-prototypes	0.850	0.845	0.848	0.852	0.488	0.746	0.410
	$\mu = 0.2 \quad \gamma = 0.7$	(0.010)	(0.011)	(0.011)	(0.011)	(0.027)	(0.013)	(0.024)
	Intuitive-K-prototypes	**0.859**	**0.862**	**0.856**	**0.861**	**0.507**	**0.755**	**0.429**
	with select prototypes	(0.002)	(0.002)	(0.003)	(0.002)	(0.009)	(0.004)	(0.003)

Table 16
The summary results for Heart disease data set(two clusters).

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Heart disease	K-prototypes [7]	0.801	0.800	0.802	0.800	0.363	0.683	0.280
	$\mu_l = 1.0$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	IK-prototypes [10]	0.812	0.804	0.809	0.808	0.390	0.700	0.308
		(0.052)	(0.050)	(0.051)	(0.051)	(0.086)	(0.044)	(0.068)
	DPC-M [30]	0.763	0.565	0.601	0.649	0.035	0.676	0.086
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.753	0.690	0.710	0.720	0.172	0.637	0.163
	$d_c = 4\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.825	0.820	0.824	0.823	0.418	0.713	0.329
	$\gamma = 0.7 \quad \alpha = 0.3$	(0.019)	(0.017)	(0.018)	(0.018)	(0.045)	(0.023)	(0.039)
	SFK-prototype [15]	0.795	0.789	0.794	0.792	0.343	0.676	0.265
	$Tr = 0.15$	(0.019)	(0.008)	(0.012)	(0.014)	(0.028)	(0.019)	(0.032)
	Intuitive-K-prototypes	**0.828**	**0.821**	**0.826**	**0.825**	**0.423**	**0.716**	**0.334**
	$\mu = 0.3 \quad \gamma = 0.2$	(0.010)	(0.014)	(0.012)	(0.012)	(0.033)	(0.015)	(0.025)
	Intuitive-K-prototypes	**0.828**	0.820	0.825	0.824	0.420	0.715	0.333
	with select prototypes	(0.010)	(0.016)	(0.013)	(0.013)	(0.035)	(0.015)	(0.027)

than the Intuitive-K-prototypes algorithm. In the Soybean data set, the Intuitive-K-prototypes algorithm achieves the best metrics.

The clustering effect comparison of mixed data sets is shown in Tables 15–18. Intuitive-K-prototypes algorithm for the ACA data set is superior to other algorithms in all average metrics and is the best in the optimal average metrics. For the heart disease data set, we consider two cases. First, for the case of two clusters, the algorithm in this paper performs well, but the gap with other algorithms is not large. The case of five clusters is a more stringent test of the clustering algorithm performance. The accuracy of the Intuitive-K-prototypes algorithm is 0.100 higher than that of the algorithm belonging to the same k partition. The minimum difference in average accuracy is 0.025. For the Chess data set, the average accuracy of the Intuitive-K-prototypes algorithm is 0.842, the optimal average value. The minimum difference in average accuracy is 0.054.

We averaged the metric values of each algorithm on different data sets. And show it in Fig. 6. The blue bar represents the average performance. The red bar represents the best performance. Compared to other algorithms, the Intuitive-K-prototypes algorithm improves the average AC by at least 4.3% and the best accuracy by at least 3.8%, improves the average FMI by at least 5.7% and the best accuracy by at least 5.6%, improves the average NMI by at least 2.8% and the best

Table 17
The summary results for Chess data set.

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Chess	K-prototypes [7]	0.737	0.735	0.737	0.736	0.224	0.614	0.169
	$\mu_l = 1.0$	(0.041)	(0.041)	(0.041)	(0.041)	(0.084)	(0.042)	(0.067)
	IK-prototypes [10]	0.794	0.785	0.788	0.789	0.333	0.672	0.263
		(0.021)	(0.016)	(0.017)	(0.018)	(0.037)	(0.020)	(0.037)
	DPC-M [30]	0.754	0.753	0.754	0.754	0.258	0.631	0.195
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.748	0.747	0.748	0.747	0.246	0.624	0.185
	$d_c = 4\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.804	0.780	0.785	0.792	0.337	0.681	0.301
	$\gamma = 0 \quad \alpha = 0$	(0.071)	(0.052)	(0.053)	(0.061)	(0.122)	(0.064)	(0.151)
	SFK-prototype [15]	0.730	0.727	0.728	0.728	0.210	0.609	0.159
	$Tr = 0.15$	(0.051)	(0.052)	(0.052)	(0.052)	(0.105)	(0.051)	(0.082)
	Intuitive-K-prototypes	**0.867**	**0.836**	**0.842**	**0.851**	**0.468**	**0.746**	**0.427**
	$\mu = 0.2 \quad \gamma = 0$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	Intuitive-K-prototypes	0.774	0.751	0.756	0.763	0.347	0.683	0.316
	with select prototypes	(0.093)	(0.085)	(0.086)	(0.089)	(0.122)	(0.063)	(0.111)

Table 18
The summary results for Heart disease data set(five clusters).

Dataset	Algorithm	PR	RE	AC	F_1	ARI	FMI	NMI
Heart disease	K-prototypes [7]	0.342	0.350	0.417	0.346	0.189	0.423	0.201
	$\mu_l = 0.8$	(0.051)	(0.059)	(0.108)	(0.051)	(0.101)	(0.093)	(0.045)
	IK-prototypes [10]	0.351	0.376	0.440	0.363	0.217	0.447	0.197
		(0.037)	(0.077)	(0.078)	(0.054)	(0.086)	(0.074)	(0.033)
	DPC-M [30]	0.320	0.304	0.311	0.308	0.095	0.341	0.205
	$d_c = 1.5\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	DP-MD-FN [31]	0.358	0.348	0.521	0.352	0.264	0.514	0.161
	$d_c = 4\%$	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	MCFCIW [12]	0.344	0.371	0.444	0.357	0.219	0.450	0.195
	$\gamma = 0.7 \quad \alpha = 0.0$	(0.051)	(0.071)	(0.084)	(0.056)	(0.106)	(0.093)	(0.062)
	SFK-prototype [15]	0.337	0.350	0.443	0.343	0.220	0.451	0.194
	$Tr = 0.15$	(0.060)	(0.067)	(0.072)	(0.050)	(0.103)	(0.087)	(0.025)
	Intuitive-K-prototypes	**0.374**	**0.360**	0.546	**0.367**	0.351	0.576	**0.241**
	$\mu = 0.3 \quad \gamma = 0.5$	(0.058)	(0.055)	(0.018)	(0.055)	(0.037)	(0.030)	(0.024)
	Intuitive-K-prototypes	0.371	0.357	**0.547**	0.365	**0.356**	**0.580**	0.240
	with select prototypes	(0.031)	(0.033)	(0.011)	(0.032)	(0.014)	(0.012)	(0.012)

accuracy by at least 5.6%, and improves the average F_1 by at least 3.0% and the best accuracy by at least 3.3%. In conclusion, the performance of the Intuitive-K-prototypes algorithm outperforms other algorithms.

6. Conclusion

This paper proposed a selecting prototypes algorithm for heuristically selecting evenly distributed initial prototypes and a clustering algorithm with intuitionistic distribution centroid and comprehensive weight strategy.

The proposed intuitionistic distribution centroid in this paper effectively utilizes both intra-cluster and inter-cluster information. It constructs the cluster center in the form of intuitionistic fuzzy sets. Based on the intuitionistic distribution centroid, the membership and non-membership of each attribute value can be more clearly distinguished, and the distance constructed on this basis positively impacts clustering. The iterative attribute weight strategy considers both the complexity within the cluster and the similarity between clusters, making the weight distribution more reasonable. Whether the data set is numerical, categorical or mixed, it can be seen from the experimental results that it can achieve satisfactory clustering performance and has less time complexity.

However, the content of this paper still has the following defects: (1) Improving algorithm adaptability. The initial prototype selection algorithm is suitable for data with a uniform distribution. Still, for data with a more complex distribution, the effect of the algorithm proposed in this paper may not be satisfactory. (2) Improve the parameter selection strategy. The selection of various parameters in this paper requires a lot of debugging experiments. It is to select suitable parameters from experience but lacks theoretical derivation.

In future work, we will focus on studying more complex data distributions and making more theoretical choices regarding parameters.

CRediT authorship contribution statement

Hongli Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Jusheng Mi:** Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jusheng Mi reports financial support was provided by National Natural Science Foundation of China. Jusheng Mi reports a relationship with the Key Project of Natural Science Foundation of Hebei province.

Data availability

Data will be made available on request.

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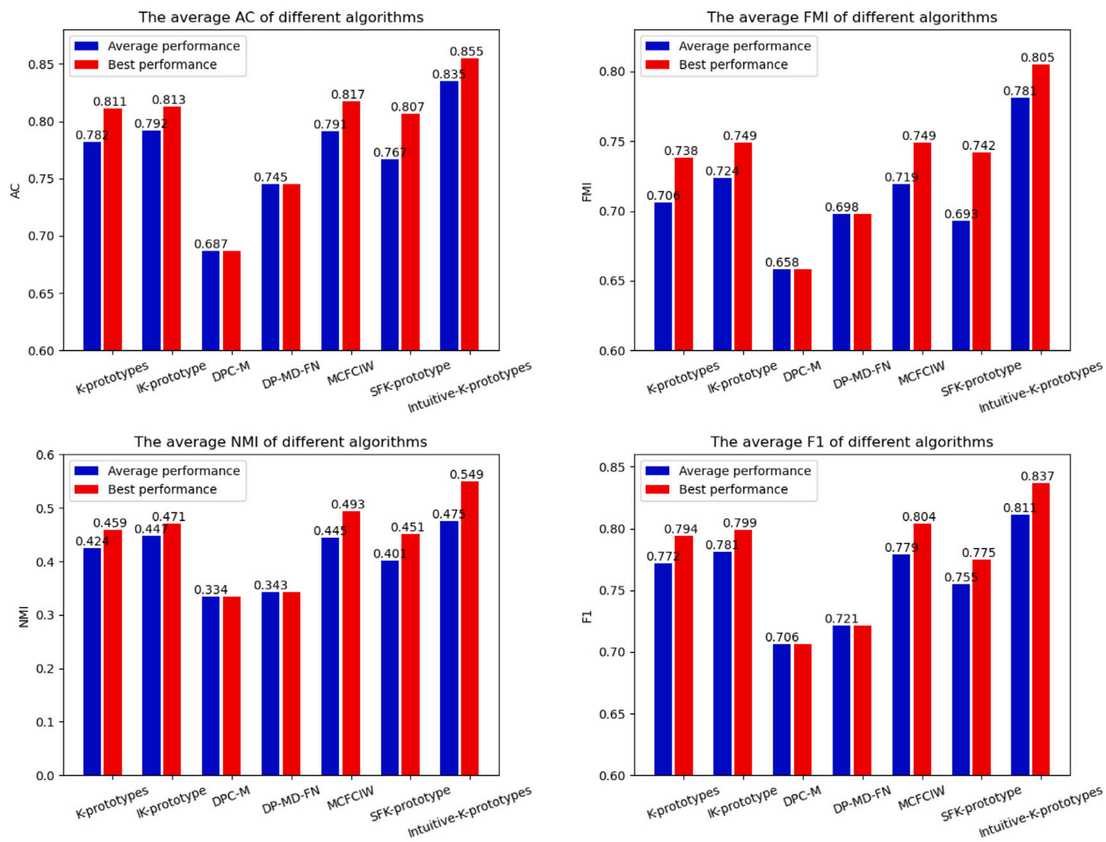

Fig. 6. Average value of the above experimental metrics. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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